

LEARNING TO COOPERATE WITH PAVLOV AN ADAPTIVE STRATEGY FOR THE ITERATED PRISONER'S DILEMMA WITH NOISE

ABSTRACT. Conflict of interest may be modeled, heuristically, by the iterated Prisoner's Dilemma game. Although several researchers have shown that the Tit-For-Tat strategy can encourage the evolution of cooperation, this strategy can never outscore any opponent and it does poorly against its clone in a noisy environment. Here we examine the family of Pavlovian strategies which adapts its play by positive and negative conditioning, much as many animals do. Mutual cooperation will evolve in a contest with Pavlov against a wide variety of opponents and in particular against its clone. And the strategy is quite stable in a noisy environment. Although this strategy cooperates and retaliates, as does Tit-For-Tat, it is not forgiving; Pavlov will exploit altruistic strategies until he is punished by mutual defection. Moreover, Pavlovian strategies are natural models for many real life conflict-of-interest encounters as well as human and computer simulations.

Keywords: game theory, evolution of cooperation, Prisoner's Dilemma, Pavlov.

1. INTRODUCTION

How can cooperation evolve when selfish or aggressive behavior has greater immediate utility? This question has been studied extensively over the past decades using simple game theoretic models. Assume two individuals meet frequently and that the payoff after each meeting is given by the Prisoner's Dilemma scheme

R, R	S, T
T, S	P, P

(see for example Rapoport and Chammah, 1965). In this paper we will concentrate on the case with $P = -1$, $R = 1$, $S = -2$, $T = 2$ so the matrix will be

$1, 1$	$-2, 2$
$2, -2$	$-1, -1$

On a single play of this famous game the 'rational' strategy is for each player to choose the second option (Defect) hoping to collect the 'temptation' payoff of $T = 2$ and avoiding the 'sucker' payoff of $S = -2$. But the resulting DD outcome penalizes each player one unit ($P = -1$) whereas if each chose the first option (Cooperate) then each would collect the reward of $R = +1$.

In iterated plays of this game the mutually cooperative strategy may prevail. Indeed many species, including mankind, have apparently learned to play the cooperative strategy in real life situations with significant similarities to this simple model. Many economists, political scientists, sociologists, zoologists and others have used the Prisoner's Dilemma as a simplified model to study how cooperation may have developed in business, politics, culture, and nature (see, for example, Axelrod, 1984; Axelrod and Dion, 1987; Dawkins, 1989; Maynard Smith, 1982; Rapoport and Chammah, 1965; and references therein). We discuss a few of these in Section 5 below.

What is an ideal strategy for playing the iterated Prisoner's Dilemma over a long and indefinite number of rounds? It should accumulate a respectable total score, it should never lend itself to exploitation, and it should be robust enough to recover after errors in the perception of outcomes.

We certainly cannot demand that a strategy score more than $+1$ on the average with the given payoff matrix. Indeed no strategy can even earn more than -1 per round when played against the strategy which always defects (All-D). Even against the Random strategy, which is really no strategy at all, the highest score that one can expect comes by choosing All-D which gives either a $+2$ or a -1 payoff for an average of $\frac{1}{2}$ per round. A strategy is called 'nice' if it is not the first to defect. One should expect that a good strategy will collect an average score close to 1 when played with nice strategies. It should do nearly as well even when played against those strategies that do not mindlessly defect for extended numbers of rounds.

The altruistic All-C strategy could clearly collect one point each round against every nice strategy, but it can be brutally exploited by the All-D strategy. We would expect a good strategy not to do much worse than any opponent for extended numbers of rounds. Any

exploitable strategy can be invaded by a mutant strategy and eventually wiped out. In the words of Maynard Smith (1982) such strategies are not evolutionarily stable.

The best known strategy for the iterated Prisoner's Dilemma, the Tit-For-Tat (TFT) strategy, requires that the player cooperates at the first meeting and thereafter echoes what her opponent has played on the previous round. In various simulations with human subjects and with computer programs playing against each other, this simple strategy has performed amazingly well. In the famous computer tournaments of Axelrod (1984), TFT scored at or close to the desired average of the reward, R , against a wide assortment of opponents. Perhaps more importantly the TFT strategy cannot be exploited. Indeed TFT can *never* do significantly worse – nor ever any better – than *any* opponent in any two person game so long as the payoff matrix is symmetric (Proposition 5).

Except in antiseptically controlled situations, random errors or noise may change an outcome or at least the perception of an outcome (Axelrod and Dion, 1988). Assume that a number of rounds of the iterated Prisoner's Dilemma have occurred between players who have tended to cooperate. At some stage the first player might misunderstand the previous play of the second or perhaps the second spontaneously alters his original strategy. A good strategy should be able to recover from this disturbance in a relatively few rounds and resume mutual cooperation against a variety of strategies including its clone.

TFT fails this test. She plays poorly against a clone of herself if, due to some random noise or misinterpretation, she believes that her opponent defected. For after the misperception she will play D to her opponent's C and on the following moves each will reverse her play. Thus they will be stuck in this CD DC CD . . . destructive echo with an average payoff of 0 per round. Another misperception is as likely to send the two into the more dreadful DD lock-in with an average payoff of -1 per round as it is to return them to mutual cooperation.

R. May (1987) suggests that strategies for the iterated Prisoner's Dilemma with some natural randomness or 'realistic stochasticity' should be explored because of the uncertainties in biological and social situations. Nowak and Sigmund (1990) echo this sentiment. 'The

answer to the opponent's last move (which may be misperceived in the first place) is never an all-or-nothing, but only an increase or decrease of the propensity to cooperate.'

In this paper we investigate a family of stochastic strategies, the Pavlovian responses $\mathcal{P}n$ for $n = 1, 2, \dots$. At each round the player will have a certain probability of cooperating. This probability will increase on the next round if he is rewarded for cooperating or punished for defecting and will decrease otherwise. The Pavlov strategy is a natural or common-sense strategy; repeat what has helped and avoid what has hurt.

In its simplest form ($n = 1$) our Pavlov will cooperate (C) *either* if he was rewarded for cooperation on the previous play, i.e., both players cooperated (CC), *or* if he was punished for defecting on the previous round, i.e., both defected (DD). If Pavlov was punished for cooperating or rewarded for defecting (CD or DC) then he will defect (D) on the next round. This strategy is significantly different from TFT. Both strategies do reinforce mutual cooperation (C follows a play of CC) and both retaliate immediately after the opponent defects (switch to D after CD). But if Pavlov defected on the previous play, he will next play the *opposite* of what TFT would on that round. For example if Pavlov defected and got away with it (DC), then he will succumb to the temptation reward of +2 and defect again on the next round rather than cooperate as would the more generous TFT. Pavlov will continue to exploit his opponent unless she too retaliates, i.e., until and unless the DD outcome occurs. Whereas TFT would continue to defect after DD, Pavlov will have been chastised by his negative score that he and, only incidentally, his opponent collected and will cooperate on the next round.

Essentially this $\mathcal{P}1$ strategy was described by Rapoport and Chammah (1965) who called it the simpleton strategy. Against the All-D strategy $\mathcal{P}1$ will alternately be punished for defecting, and so switch to cooperation, and then be punished even more for cooperating and so switch back to defection. His expected payoff will thus be the average of -1 and -2, i.e., -1.5, while the opponent will collect the average of -1 and +2, i.e., 0.5. We do not consider $\mathcal{P}1$ to be a particularly commendable strategy.

In Pavlov and the Prisoner's Dilemma (Kraines and Kraines 1989),

hereafter referred to as PPD, we studied the slow learning, or perhaps the less impetuous, variant of this strategy with $n = 10$. Initially $\mathcal{P}n$, or more precisely the sub strategy $\mathcal{P}(k, n)$, has a certain probability k/n of cooperating. This probability of cooperating will increase by $1/n$ both after a CC outcome (reward of $+1$ for cooperation) and after a DD outcome (punishment of -1 for defecting). It will decrease by $2/n$ after a CD or DC outcome (punish for cooperating and getting the sucker payoff of -2 or reward for defecting and winning the temptation payoff $+2$). This prescription must be slightly modified to ensure that the probability of cooperating remains between 0 and 1. See Definition 1 below.

The strategy evolves in response not only to the opponent's action, as in TFT and many other strategies, but also to the success or failure of its own action on previous rounds. As a result these strategies learn to cooperate with each other and also are stable under perturbations. The strategies do not require a rational player or even one with any foresight at all. Pavlov adapts his strategy, as did I.P. Pavlov's dogs, merely to positive and negative conditioning. He does not need to be aware of what his opponent plays or wins or loses, or even that a game is being played, in order to learn to cooperate in encounters modeled by the iterated Prisoner's Dilemma.

Yet Pavlov is not strictly an automaton. The previous outcomes do not determine the actual response of Pavlov but only the likelihood of a particular response. Two identical strategies would in general evolve quite differently over the iterated plays, although in many cases they would have the same limiting behavior.

Although some opponents, such as All-D, can exploit $\mathcal{P}n$, the initially cooperating Pavlov strategy $\mathcal{P}(n, n)$ is frequently superior to TFT. Both will cooperate with nice opponents, in particular with their clones. Pavlov rarely does significantly worse, and occasionally does much better, than his opponents. In particular, Pavlov will exploit the All-C and the Random strategies. And in those cases where Pavlov does worse, the opponent also suffers in this non-zero sum game.

We substantiate these claims quantitatively by analyzing how the Pavlov strategy performs against a wide variety of other strategies. See Appendix A for a list of these alternative strategies. More specifically we compute the expected winnings for Pavlov and for opponent

strategies in iterated games with and without a fixed probability for ending (the discount parameter). In Section 4 we show that the best variant of the Pavlovian strategies is $\mathcal{P}n$ for $n = 3$ or 4 . In Section 5 we discuss a few ‘real life’ situations in the literature which have been modeled by the iterated Prisoner’s Dilemma and show that the Pavlov strategy fits the actual responses better than the TFT strategy.

2. FORMAL DEFINITIONS

In this section we make the Pavlov strategy precise and compare it with other strategies that have been proposed in the literature. Technically for a fixed n the strategy $\mathcal{P}n$ is a family of mixed strategies $0 < k < n$: $\mathcal{P}(k, n) = \text{cooperate with probability } k/n$. We make the convention that $\mathcal{P}(k, n) = \mathcal{P}(0, n)$ if $k < 0$ (certainly defect) and $\mathcal{P}(k, n) = \mathcal{P}(n, n)$ if $k > n$ (certainly cooperate). This mixed strategy is updated at the end of each round in the following way. Assume that $m - 1$ rounds have been played against strategy $\mathcal{S}$ and that Pavlov uses strategy $\mathcal{P}(k, n)$ on round m . Then there is a certain probability p with which $\mathcal{S}$ will cooperate (p might depend on the outcomes of the previous $m - 1$ rounds, on Pavlov’s current strategy, and/or on a random process). Thus the outcome of round m will be CC with probability kp/n and will be DD with probability $(1 - k/n)(1 - p)$. With either of these outcomes, the strategy $\mathcal{P}(k, n)$ will transform to $\mathcal{P}(k + 1, n)$, meaning that the player will be somewhat more likely to cooperate on the following play. A CD or DC outcome means that either Pavlov was rewarded 2 for defecting or punished 2 for cooperating. In either of these events, his propensity for cooperation will decrease to $\mathcal{P}(k - 2, n)$.

More formally:

DEFINITION 1. The operator $\mathbb{A} = \mathbb{A}_m(\mathcal{P}n, \mathcal{S})$ given by

$$\mathbb{A}\mathcal{P}(k, n) = \begin{cases} \mathcal{P}(k + 1, n) \text{ with probability } \frac{k}{n}p + \left(1 - \frac{k}{n}\right)(1 - p) = \frac{(2k - n)p + n - k}{n} \\ \mathcal{P}(k - 2, n) \text{ with probability } 1 - \frac{(2k - n)p + n - k}{n} = \frac{k - (2k - n)p}{n} \end{cases}$$

defines the transition of Pavlov's strategy from round $m - 1$ to round m .

For example, if $\mathcal{S}$ is All-D, and so $p \equiv 0$, the transition probabilities are $1 - k/n$ to increase cooperation one step and k/n to decrease it two steps. If $\mathcal{S}$ is TFT then p is 1 if Pavlov actually cooperated on the previous round and 0 otherwise.

Assume that Pavlov starts with the mixed strategy $\mathcal{P}(5, 10)$: cooperate with probability 50%. If the outcome of the first play is either CC or DD, then Pavlov will cooperate with 60% probability on the second play, i.e., Pavlov will use mixed strategy $\mathcal{P}(6, 10)$. If the outcome were CD or DC he would cooperate with only 30% probability on that round, i.e., use mixed strategy $\mathcal{P}(3, 10)$ on the next round. If Pavlov were to start with $\mathcal{P}(0, 10)$, the pure strategy of defect with certainty, and if $\mathcal{S}$ were also to defect, then $\mathbb{A}\mathcal{P}(0, 10) = \mathcal{P}(1, 10)$ while if $\mathcal{S}$ were to cooperate then $\mathbb{A}\mathcal{P}(0, 10) = \mathcal{P}(0, 10)$ since negative probabilities do not exit.

The operator $\mathbb{A}$ above will either increase the likelihood of cooperation by $1/n$ or decrease this likelihood by $2/n$. These fractions are dependent on the particular entries in the payoff matrix; if the payoff matrix were

R, R	S, T
T, S	P, P

for integers R, S, T , and P , then the formula for $\mathbb{A}\mathcal{P}(k, n)$ must be modified so that it will equal $\mathcal{P}(k + R, n)$ with probability kp/n corresponding to the CC outcome. The strategy could also transform to $\mathcal{P}(k + S, n)$, $\mathcal{P}(k - T, n)$ or $\mathcal{P}(k - P, n)$ with appropriate probabilities and subject to the obvious boundary constraints.

A very similar strategy was studied independently by Macy (1989, 1991). Macy assumes that each player has a probability of cooperation p at each stage and that this probability is then modified by the reinforcement for cooperation and equals $p + a(1 - \lambda^\delta)$ where a represents the payoff, $R, S, -T$ or $-P$, received by the player on the previous round, λ is either the original probability p if the payoff a is greater than 0 or it is $1 - p$ if $a < 0$, and δ is an adjustable constant. One disadvantage of this model is that it assumes a continuum of mixed strategies and is thus not amenable to study by Markov chains.

Assume that Pavlov is in state $\mathcal{P}(k, n)$ and that $\mathcal{S}$ will cooperate with probability p on the next round. Then Pavlov can expect to receive 1 unit if both cooperate (probability kp/n), 2 units if he defects while $\mathcal{S}$ cooperates, etc. Thus the expected payoff to Pavlov is

$$\begin{aligned} \text{Py}\mathcal{P} &= \frac{k}{n} p + 2\left(1 - \frac{k}{n}\right)p - 2\frac{k}{n}(1-p) - 1\left(1 - \frac{k}{n}\right)(1-p) \\ &= -1 - \frac{k}{n} + 3p \end{aligned} \tag{1}$$

Similarly the expected payoff to $\mathcal{S}$ is

$$\begin{aligned} \text{Py}\mathcal{S} &= \frac{k}{n} p - 2\left(1 - \frac{k}{n}\right)p + 2\frac{k}{n}(1-p) - 1\left(1 - \frac{k}{n}\right)(1-p) \\ &= -1 + 3\frac{k}{n} p \end{aligned} \tag{2}$$

For example assume $n = 3$ and $\mathcal{S}$ is All-D (so $p = 0$). To compute the expected payoff, note that from state $\mathcal{P}(1, 3)$ Pavlov will cooperate and collect -2 with probability $1/3$ and defect and collect -1 with probability $2/3$. Thus starting from that state Pavlov can expect to collect $-2/3 - 2/3 = -4/3$. We combine the expected payoffs for the other states into the *payoff column vectors*

$$\begin{aligned} \mathbb{P} &= \left[-1, -\frac{4}{3}, -\frac{5}{3}, -2\right]^T \quad \text{for Pavlov and} \\ \mathbb{S} &= \left[-1, 0, 1, 2\right]^T \quad \text{for All-D.} \end{aligned} \tag{3}$$

We now formalize the notion of an ideal strategy outlined in Section 1.

DEFINITION 2. An *ideal strategy* for the iterated Prisoner's Dilemma is one that satisfies the criteria below.

1. The average score per round should approach $+1$ as the number of rounds increases, at least against a wide variety of other strategies (it does well for itself among friends).
2. Over many plays, the average of the differences in payoffs between

this strategy and every opponent should approach a number greater than or equal to 0 (it is not exploitable by anyone).

3. It will quickly recover from perceived defections due to noise or small perturbations, at least against its clone (it learns, and relearns, to cooperate).

An ideal strategy will overcome past fights and disagreements, no matter who was at fault. Perhaps this situation is somewhat reminiscent of children who could be bitter rivals and always fighting when young but grow up to be mutually supportive. This example motivates the following definition.

DEFINITION 3. Two strategies are called *fraternal* if, given any initial sequence of outcomes, there is a number q such that after q additional rounds each will average close to 1 per round. The expected value of q is called the *training time*. A *clan* is a set of strategies $\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_s$ such that each strategy is fraternal with all other strategies of the set as well as with a clone of itself.

Note that since fraternal strategies eventually average close to the same 1 unit per round, neither will score many more points than the other. And by definition they will restore mutual cooperation after noise or perturbations. Thus all strategies in a clan satisfy the conditions for an ideal strategy *provided* that all the opponents are restricted to members of the clan.

The All-C strategy and TFT are fraternal but as we saw above TFT is not fraternal with another copy of TFT and so All-C and TFT do not form a clan. The variant, Tit-for-Two-Tats (TF2T), which cooperates unless the opposing strategy defects twice in succession, is also not fraternal with its clone for if each were to defect twice in a row, or perceive that the other did so, then each would continue to defect.

Molander (1985) and Mueller (1987) each propose modifications of the TFT strategy which will not retaliate with certainty nor always defect after a DD outcome. These are called the Error Forgiving ($\mathcal{EF}$) strategy and the Conditional Cooperating ($\mathcal{CC}$) strategy respectively. But unless the adjustable parameters for not defecting in those strategies are large, in which case these strategies are exploitable, the

recovery or training time will be long. The set of strategies $\{\text{All-C}, \mathcal{EF}, \mathcal{CC}\}$ can be shown to form a clan, but its members are each exploitable by strategies outside the clan and so we do not consider these to be close to ideal.

A more sophisticated stochastic variation of TFT has been proposed by Nowak and Sigmund (1989, 1990). This *reactive strategy*, $\mathcal{NS}$, will cooperate or defect with a certain probability depending on the play of the opponent on the previous round. More precisely consider the family of strategies $\mathcal{NS}(y, p, q)$ determined by a triple (y, p, q) where y represents the probability of cooperating initially, p is the probability of cooperating if the opponent cooperated on the previous move and q is the probability of cooperating if the opponent defected on the previous move. Extreme cases of this strategy include TFT ($p = 1$ and $q = 0$), All-D ($p = q = 0$) and All-C ($p = q = 1$). The $\mathcal{NS}(y, p, q)$ strategy is asymptotically cooperative with its clone if and only if $p = 1$ and $q > 0$. Nowak and Sigmund give formulas for the expected payoffs between clones in some generality. With our payoff matrix, if $p = 2/3$ and $q = 1/3$ then the expected long term average payoff in a match with its clone will be 0. The Nowak–Sigmund strategy is quite different from the Pavlov strategy. The probabilities for cooperation by $\mathcal{NS}$ can assume only two values rather than $n + 1$ values. More importantly the choice of probabilities in their strategy depend on the opponent's move rather than on the player's reward or punishment.

The Pavlov strategy has many similarities to evolutionary game theory models as described in Mailath (1992) and the references therein. Instead of a single player, $\mathcal{P}(k, n)$, who has a k/n propensity to cooperate, these models often assume that there is a group of n individuals of whom k will certainly cooperate while the others will certainly defect. Call this cumulative strategy $\mathcal{E}(k, n)$. The individuals who receive a positive payoff are more likely to reproduce effectively in the next generation while those who receive a negative payoff are more likely to die off. For n very large, there is a number u , possibly negative, depending on k and the payoff matrix and perhaps other factors so, in the next generation, the cumulative strategy will change to $\mathcal{E}(k + u, n)$. In other words, the population size will remain the same but the number of cooperators will increase if $u > 0$ or decrease if $u < 0$.

The evolutionary model is more deterministic than the Pavlov strategy. In particular, if a population has been selected for cooperation, i.e., the cumulative strategy is $\mathcal{E}(n, n)$ at some stage, then presumably that population can never change or at most change very slowly while the $\mathcal{P}(n, n)$ strategy changes immediately to $\mathcal{P}(n, n - 2)$ if he receives the sucker payoff. Evolutionary models generally have a large population or magnitude for n . In Section 4 we show the Pavlov strategy is more effective when the value of n is of the order of 3 or 4, since for larger n the learning time is extraordinarily long.

3. PAVLOV AGAINST OTHER STRATEGIES

How does strategy $\mathcal{P}n$ fare in head-to-head matches against various opponents in the iterated Prisoner's Dilemma with a large and indefinite number of rounds? We can analyze this question mathematically using Markov chains to find under what circumstances the players will eventually stabilize at mutual cooperation (become trained) and what is the expected time to train. Additionally we compute the expected payoff to each player. The tables and figures are generated with the powerful matrix algebra program, 386-MATLAB (1990). The MATLAB programs we have written are available on request.

We consider opponents such as the All-D or meanie strategy, various random strategies, the All-C or altruistic strategy, the TFT strategy, and the Pavlov strategy itself. We also examine intelligent strategies (the Perfect Trainer and Normal Trainer) which lead to mutual cooperation against $\mathcal{P}n$ quite rapidly. Appendix A contains a list of strategies considered in this paper. Many of the details will be relegated to the other appendices. See also PPD for more background.

The Markov transition matrix for $\mathcal{P}n$ versus $\mathcal{S}$ corresponds to the transformation $\mathbb{A} = \mathbb{A}(\mathcal{P}n, \mathcal{S})$ of Definition 1. The states in the Markov chain are the $(n + 1)$ allowable probabilities of cooperation for a given round, $0/n, 1/n, 2/n, \dots, n/n$, coupled with the given opponent's strategies for that round. That is, the states catalog the possible inputs to the next play. This formulation is quite different from that used in Chapter 7 of Rapoport and Chammah (1965). The states in Markov matrices in that source and several others in the game theory

literature are related to the four possible outcomes (CC, CD, DC, and DD) rather than the mixed strategies, or inputs, as above.

In PPD, we showed that the Markov chain $\mathbb{A} = \mathbb{A}(\mathcal{P}n, \text{All-D})$ is a *regular* transformation matrix, meaning that for any initial state there is a positive probability of transforming into any other state after a sufficiently large number of rounds. For such a chain each row of $\mathbb{A}^t$ will approach the same *equilibrium state vector* as t increases. This vector can be computed as the normalized (left) eigenvector of the matrix corresponding to eigenvalue 1. As n increases, $\mathcal{P}n$ tends to use strategies nearer to $\mathcal{P}(n/3, n)$ with less and less variance. More precisely the coordinates of equilibrium state vector will be very small if k/n is not close to $1/3$ (see Figure 1 in PPD). From $\mathcal{P}(n/3, n)$, Pavlov can expect to lose 2 with probability $1/3$ and 1 with probability $2/3$ for an average loss of $4/3$. On the other hand, All-D can expect to gain 2 with probability $1/3$ and lose 1 with probability $2/3$ and thus just break even. The slower learning variety of Pavlov will outscore the faster learning variety by a small amount. The TFT strategy is a somewhat better strategy than $\mathcal{P}n$ against All-D; it will collect the constant payoff of -1 on each play after the first round instead of -1.33 .

For fixed p between 0 and 1, the strategy $\mathcal{C}p$ will cooperate with probability p regardless of the previous history. If $p = 1$, this is the All-C strategy while if $p = 0$ it is the All-D strategy. For arbitrary p , this strategy coincides with the Nowak–Sigmund strategy $\mathcal{NS}(p, p, p)$. We call the special case with $p = \frac{1}{2}$ the Random strategy. If $\mathcal{P}1$ plays against the $\mathcal{C}\frac{1}{2}$ or Random strategy, then by definition all outcomes are equally likely on the average so the expected payoff will be $1 - 2 + 2 - 1 = 0$ exactly as in the case of TFT versus Random. But using eigenvector methods, we can see that, for larger n , Pavlov exploits Random. For example with $n = 6$ Pavlov will tend to defect more often than he cooperates and collect an average payoff of about 0.27 per play to Random's -0.8 . Indeed, for large n , the expected payoff to Pavlov and to Random can be shown to be asymptotic to

$$\begin{aligned} \frac{1}{2} - \frac{1}{n} \frac{\sqrt{5} + 1}{2} &\approx 0.5 - \frac{1.618}{n} \quad \text{and} \\ -\frac{3}{2} + \frac{3}{n} \frac{\sqrt{5} + 1}{2} &\approx -1.5 + \frac{4.854}{n}, \quad \text{respectively.} \end{aligned} \quad (4)$$

The slower learning Pavlov again collects more than his faster learning cousins. In contrast with the previous match, Pavlov reaps more reward than TFT would have against Random.

If $p < 1$ then the Markov chain for a match of $\mathcal{P}n$ against $\mathcal{C}p$ will also be regular and so there will be an equilibrium state vector for Pavlov depending on n and p . In Table I we list the expected payoffs to $\mathcal{P}(10, 10)$ and $\mathcal{C}p$ for values of p in steps of 0.1 from 0 to 1. Note that $\mathcal{C}p$ outscores $\mathcal{P}(10, 10)$ only if $p < 0.3$. But in those cases $\mathcal{C}p$ will also lose unless $p = 0$, i.e. except against the All-D strategy. In Table II we list the values of p for various n at which each opponent loses the same amount. These values asymptotically approach about 0.2. Pavlov will exploit all but the very uncooperative random strategies.

Similar results hold for the Nowak–Sigmund strategy. Unless the rates of cooperation, p and q , are very low, $\mathcal{NS}(y, p, q)$ will be outscores in a match against Pavlov. And if these initial probabilities are low, then both players will lose in this non-zero sum game.

Against All-C, the $\mathcal{P}1$ version of Pavlov will continue to cooperate if he starts in that mode, i.e., using strategy $\mathcal{P}(1, 1)$, and each will collect a 1 unit payoff per round. But if Pavlov starts out defecting, i.e., using strategy $\mathcal{P}(0, 1)$, then he will continue to defect (exploiting All-C) and so win +2 each round while his opponent loses 2. As with his performance against Random, Pavlov is not benevolent or altruistic and will occasionally collect a higher payoff than TFT would against some strategies.

TABLE I
Payoffs for $\mathcal{C}p$ versus $\mathcal{P}(10, 10)$.

p	Pavlov	$\mathcal{C}p$
0	-1.3370	0.0109
0.1000	-1.0089	-0.1732
0.2000	-0.6761	-0.3717
0.3000	-0.3389	-0.5834
0.4000	0.0015	-0.8046
0.5000	0.3431	-1.0293
0.6000	0.6834	-1.2501
0.7000	1.0199	-1.4598
0.8000	1.3511	-1.6532
0.9000	1.6761	-1.8284
1.0000	1.0000	1.0000

TABLE II

$\mathcal{C}p$ versus $\mathcal{P}(n, n)$; values of p for which payoffs to the two players are equal, and the payoff to the two players

n	p	payoffs
1.0000	0.5000	0.0000
2.0000	0.3819	-0.2361
3.0000	0.3497	-0.3005
4.0000	0.3218	-0.3565
5.0000	0.3033	-0.3934
6.0000	0.2888	-0.4225
7.0000	0.2775	-0.4449
8.0000	0.2686	-0.4628
9.0000	0.2614	-0.4772
10.0000	0.2555	-0.4891
11.0000	0.2505	-0.4990
12.0000	0.2464	-0.5073
13.0000	0.2428	-0.5144
14.0000	0.2398	-0.5204
15.0000	0.2371	-0.5257
16.0000	0.2348	-0.5304
17.0000	0.2328	-0.5344
18.0000	0.2310	-0.5380
19.0000	0.2294	-0.5413
20.0000	0.2279	-0.5441

For larger n , $\mathcal{P}n$ is much more likely to end up exploiting All-C than cooperating with that strategy. For example $\mathcal{P}(2, 4)$ will transform to $\mathcal{P}(3, 4)$ half the time (CC) and to the absorbing exploitative state, $\mathcal{P}(0, 4)$, the other half (DC). Although Pavlov is more likely to move from state $\mathcal{P}(3, 4)$ to the absorbing mutual assured cooperation state, with 25% probability he will fall to $\mathcal{P}(1, 4)$ and from there most likely to $\mathcal{P}(0, 4)$. In Appendix B we show that the probability that $\mathcal{P}(2, 4)$ will end up in the exploitive state, $\mathcal{P}(0, 4)$, is 61%. Similarly $\mathcal{P}(5, 10)$ will transform, in an average of five rounds, to one of the two stable states. With about 76% probability that state will be the DC state in which Pavlov exploits All-C unmercifully. Pavlov thus usually builds up a big advantage over All-C (increasing the difference by 4 points per round) while TFT will play evenly at +1 per round. Pavlov is an exploitive, rather than cooperative, strategy in the sense of Orkin (1987).

If an intelligent opponent (our Perfect Trainer or PT) knows the probability with which Pavlov will cooperate on the next round, then $\mathcal{P}(k, n)$ can be coaxed into the absorbing state of mutual cooperation quite rapidly. We assume only that PT knows Pavlov's mixed strategy at each round but not that she knows the outcome of the random process which Pavlov uses, consciously or not, to decide whether he cooperates. Her strategy is simply to cooperate if she knows that Pavlov will do so with probability greater than 0.5 and defect otherwise. With this strategy $\mathcal{P}(0, 1)$ becomes trained on the second round since the first outcome is necessarily DD. For larger n the training time increases, but by a relatively modest amount. About 40 iterations are needed to transform $\mathcal{P}(5, 10)$ to $\mathcal{P}(10, 10)$. During this training process, PT can be expected to collect an average of 7 more units than Pavlov.

PT attempts to punish an opponent whom she expects to be bad and reward one she expects to be good. PT will occasionally err. If Pavlov is in state $\mathcal{P}(4, 10)$, then there is a 40% chance that Pavlov will cooperate and PT will wrongly punish him. Similarly if Pavlov is in state $\mathcal{P}(6, 10)$ there is a 40% chance that PT will reward a bad opponent not only reinforcing his bad behavior but also hurting herself. Yet in PPD we showed that this strategy is the most efficient way to train Pavlov. See Figure 3 for comparisons of training times.

One might guess that after Pavlov is in the trained state, $\mathcal{P}(n, n)$, PT can gain an extra advantage by sneaking in an occasional defection, collecting the temptation payoff of +2 rather than CC reward of +1, and then quickly 'retraining' Pavlov. But a closer examination shows that the expected cost of retraining would more than offset the immediate advantage gained unless Pavlov is a very slow learner. For example after a defection against $\mathcal{P}(1, 1)$, Pavlov will certainly defect so PT should also defect and each will collect -1. Although Pavlov will then be trained again, in these two games PT has gained a total of +1 rather than the +2 she would have had if she did not cheat.

Against strategy $\mathcal{P}(10, 10)$, PT can get away with a probing defect only for three rounds on the average. The initial defection will still leave Pavlov in the mood to cooperate with 80% probability. If both were to cooperate, and so each receive +1, then Pavlov will cooperate again with 90% probability. Thus with 72% probability, PT would

have retrained Pavlov in two plays and would be one unit ahead of what she would have collected had she always played C. But with 28% probability she will lose her advantage and much more. The defection by PT will drop Pavlov to the state of 8/10 cooperation. In Appendix C we show that PT can expect an average of 0.92 units on the first two rounds and only 1.676 units in three rounds after the defection. Thus for the four rounds she would receive 3.676 instead of the 4 she would have had by not cheating.

DEFINITION 4. The deficit after t rounds in a match between Pavlov starting at $\mathcal{P}(k, n)$ and strategy $\mathcal{S}$ is the difference between the amount the Perfect Trainer would have collected and the amount that strategy $\mathcal{S}$ collects during the next t rounds. A negative deficit is called a *surplus*.

Thus the cheating PT has a surplus of 1 against $\mathcal{P}(n, n)$ for $t = 1$. Since PT will eventually retrain Pavlov to cooperate, for each n the deficits for the cheating PT approach a limit as the length of the match increases. The deficit for the cheating PT versus $\mathcal{P}(10, 10)$ is $0.24 = 4 - 3.76$ for $t = 4$ rounds and approaches a limit of 7.9 as $t \rightarrow \infty$. This means that in the long run, PT would do almost 8 units worse in the long run by cheating just once than had she correctly played her strategy.

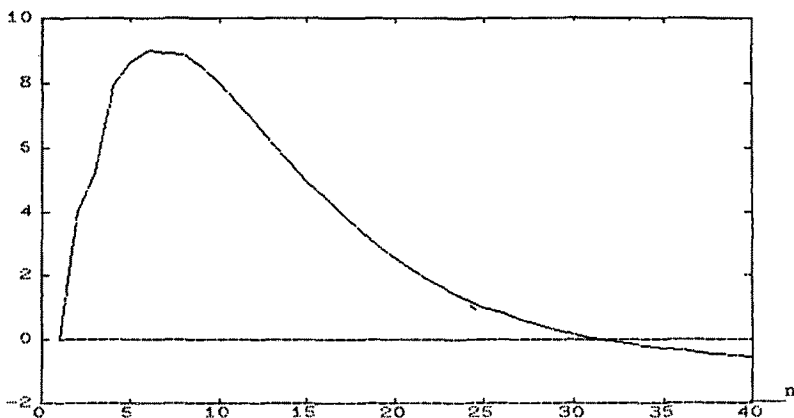


Fig. 1. Cheating Perfect Trainer versus $\mathcal{P}(n, n)$. Deficit as a function of n .

Figure 1 shows a plot of the deficits for various values of n . Note that for $n > 31$, PT will actually maintain a small surplus (or negative deficit) forever; PT can get away with a cheat if Pavlov is sufficiently slow learning. See Appendix C for details.

The PT strategy – i.e., cooperate with opponents likely to cooperate and defect otherwise – is an excellent one not only against $\mathcal{P}n$ but also against most other strategies. But one can construct strategies that exploit PT viciously. Against the strategy $\mathcal{C}0.51$, which plays C with probability 51%, our formula demands that PT should always cooperate. Her average score will be $0.51 - 2 \cdot 0.49 = -0.47$ units per round while an opponent who defected at each play would have collected $2 \cdot 0.51 - 0.49 = +0.53$. For comparison, $\mathcal{C}0.51$ will collect $0.51 + 2 \cdot 0.49 = 1.49$ against this not all that wise PT.

Even when dealing with simplified game theoretic models, it is unrealistic to imagine that any opponent could know the exact state of mind of Pavlov. If instead, on the basis of careful analysis of past performance, the intelligent but not omniscient opponent (our Normal Trainer or NT) only guesses at the probability state of Pavlov, then training can be accomplished in somewhat more time than with our inimitable PT strategy but still quite rapidly.

Assume that $\mathcal{P}n$ were actually in state $\mathcal{P}(k, n)$ and so would cooperate with probability k/n . In PPD we postulated that the probability that NT would guess that Pavlov is at least 50% likely to cooperate is k/n . For example, if $k = 0$ then we assume that NT would know with certainty that the probability that Pavlov will defect is at least $\frac{1}{2}$ and so NT will certainly defect. But if $k = 6$ and $n = 10$, then NT only has a 60% probability of correctly guessing that Pavlov will more likely cooperate than defect and so would cooperate herself with probability 0.6.

In effect NT's strategy is to cooperate with exactly the probability with which Pavlov will cooperate. If Pavlov is initially selfish then NT should be aggressive and punishing as well, even though it may hurt her as much as her opponent. When Pavlov shows more signs of cooperation, NT should reward and encourage this tendency. Her strategy is first the stick and then the carrot.

A strategy submitted to the Axelrod computer tournament by L. Downing estimated whether the opponent will more likely cooperate

or defect and then did likewise. Since the rules for estimation were necessarily imperfect, this strategy is comparable to NT. The Downing strategy and the variant, C-Downing, performed especially well against

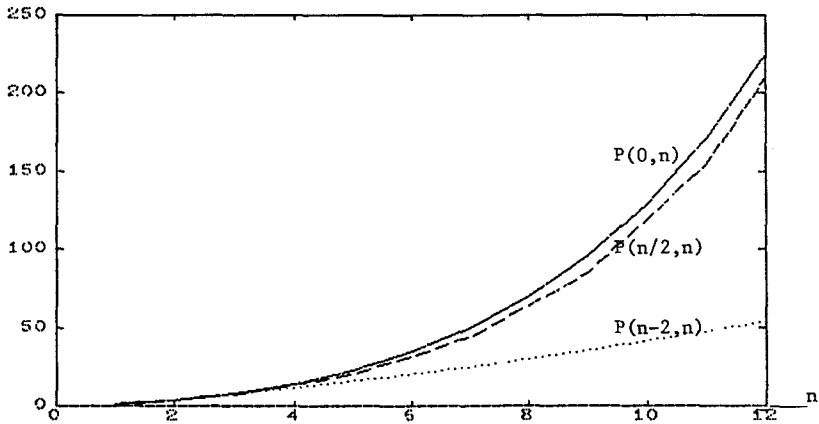


Fig. 2. Normal Trainer versus $P(0, n)$, $P(n/2, n)$ and $P(n-2, n)$. Training time as a function of n .

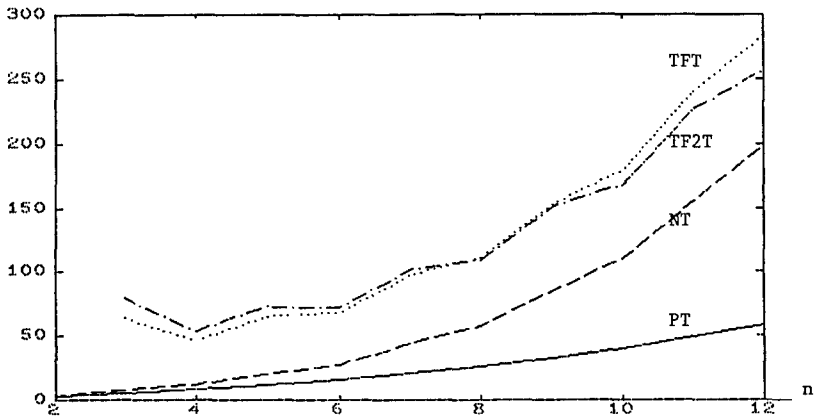


Fig. 3a. $P(n/2, n)$ versus Perfect Trainer, Normal Trainer, TFF2T and TFF2T. Training time as a function of n .

deterministic strategies such as TFT and TF2T and it would also do well against All-C, All-D and $\mathcal{P}n$. In a follow-up computer tournament run by Donninger (1986), C-Downing actually won the round robin contests under the assumption that the payoff were dynamic or that there was random noise. We feel that strategies such as C-Downing and Normal Trainer are as close as possible to an ideal strategy in the sense of Definition 1.

Note that NT cannot be exploited by $\mathcal{C}0.51$. The chance that NT will cooperate is only 51% so her expected score is $0.51^2 - 0.49^2 + 2 \cdot 0.49 \cdot 0.51 - 2 \cdot 0.51 \cdot 0.49 = 0.02$, exactly the same as her opponent. But although NT does not suffer against $\mathcal{C}0.51$, Pavlov exploits $\mathcal{C}0.51$ even more than Random.

In Figure 2 we compare the training times for NT against $\mathcal{P}(0, n)$, $\mathcal{P}(n/2, n)$, and $\mathcal{P}(n-2, n)$ for various values of n . Note that the initial cooperation probability of Pavlov has little affect on the training time unless it is very close to 1. In Figure 3a we plot the training time for the average Pavlov, $\mathcal{P}(n/2, n)$, versus NT and several other opponents. The semilog plot in Figure 3b indicates that the training time is approximately exponential in n for all but the insightful PT strategy.

As we have noted above, Pavlov sometimes does better and sometimes does worse than his opponent. In contrast TFT can never

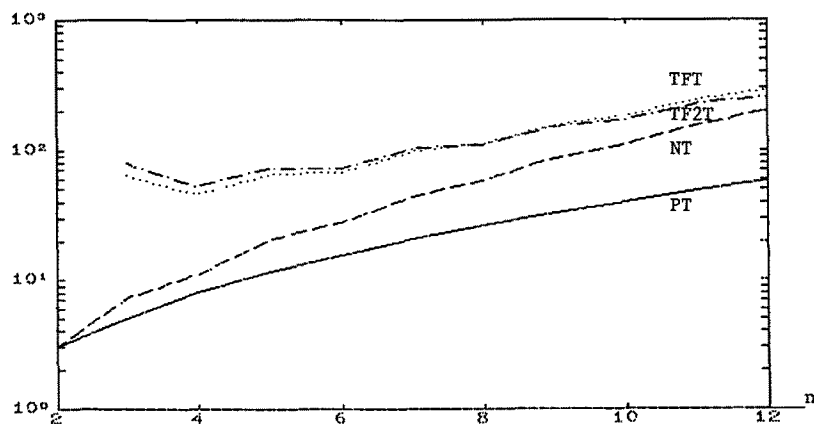


Fig. 3b. Semi-log plot of Figure 3a.

outscore any strategy, nor can she possibly lose much. This result is true provided only that the payoff matrix is symmetric. Compare p. 220 in Dawkins (1989) and p. 263 in Nowak and Sigmund (1990).

PROPOSITION 5. *Consider any 2×2 game (not necessarily satisfying the ordinal relations for a Prisoner's Dilemma) with symmetric payoff to the players given by*

$$\begin{array}{cc} R, R & S, T \\ T, S & P, P \end{array}.$$

Assume that the first player uses the TFT strategy against any second strategy $\mathcal{S}$. Then the cumulative payoff to TFT after a given number of plays will be exactly the same as that to $\mathcal{S}$ provided only that $\mathcal{S}$ had cooperated on the preceding round. It will be exactly $T - S$ units lower if $\mathcal{S}$ had defected on that round. Similarly the Suspicious TFT strategy (initially defect and thereafter play TFT) will receive exactly the same payoff as $\mathcal{S}$ or exactly $T - S$ units more depending only on whether $\mathcal{S}$ defected or cooperated on the previous round.

Proof. On the first round TFT will cooperate. If $\mathcal{S}$ also cooperates then each will collect the payoff of R and so the difference between the payoffs remains 0. This equality of payoffs continues until $\mathcal{S}$ defects. The first time that $\mathcal{S}$ defects he collects the temptation payoff T while TFT collects the sucker payoff S so the difference in winnings becomes $T - S$. This advantage of $\mathcal{S}$ over TFT will continue for as long as $\mathcal{S}$ defects because TFT will likewise defect resulting in payoffs of P to each. When $\mathcal{S}$ resumes cooperation he loses his advantage to the still defecting TFT and the two players will again be even in their earnings. The situation continues in this pattern. The result for Suspicious TFT is entirely analogous. ■

In the Prisoner's Dilemma version that we have been considering, $T = 2$ and $S = -2$, and so this difference is $+4$. Thus TFT will either have the same winnings as her opponent or will lag exactly 4 units behind. In Axelrod's computer tournament with payoff matrix

$$\begin{array}{cc} 3, 3 & 5, 0 \\ 0, 5 & 1, 1 \end{array}$$

the difference in winnings between TFT and her opponent was frequently either 0 or 5 ($= T - S$) points lower (compare the first column with the first row of Table 3, p. 194, Axelrod, 1984). The other differences in that table are integers between 0 and 5 and presumably occur because scores were averaged over five runs. (A programming error caused some of the entries in Table 8 in PPD to be fractionally off.)

For $n = 1$ or 2, Pavlov and TFT can get locked into a destructive cycle much as we saw with TFT against TFT. Neither $\mathcal{P}(1, 1)$ nor TFT are the first to defect so each will collect +1 per round, but if Pavlov started with strategy $\mathcal{P}(0, 1)$ against TFT then both will lose substantially. Since TFT will initially cooperate while Pavlov defects (DC), on the next round Pavlov will defect again because of his temptation reward and TFT will retaliate resulting in the DD outcome. On the third round Pavlov will switch to cooperation, be punished for his cooperation (CD) and so revert to defection just as TFT switches (one round too late) to cooperation (DC). This cycle continues forever. The payoffs that Pavlov receives are +2, -1, -2, +2, ... with an average of $-\frac{1}{3}$ per round. As one can see directly or by Proposition 5, TFT will have the same average payoff over the long run.

A match between $\mathcal{P}2$ and TFT has six possible states in the Markov matrix of mixed strategies: D0, C0, $D\frac{1}{2}$, $C\frac{1}{2}$, D1 and C1. The state C1 means that Pavlov cooperated on the previous round *and* will cooperate on the current one. This absorbing state of mutual cooperation can only be reached (and then only with probability $\frac{1}{2}$) through the state $C\frac{1}{2}$. Pavlov cooperated on the previous round and will do so with 50% probability this round. From any other initial strategy, states cycle endlessly; the Markov chain for $\mathcal{P}2$ against TFT is not an absorbing chain even though mutual cooperation is an absorbing state. Once in this cycle, the average payoff to each will be $-\frac{3}{7}$ over the long run. We represent the state transformations graphically below where a double arrow means certain transition and a single arrow represents a 50-50 chance of transition.

$$\begin{array}{c}
 C0 \Rightarrow D0 \leftarrow C\frac{1}{2} \rightarrow C1 \\
 \uparrow \quad \nwarrow \quad \searrow \\
 D1 \leftarrow D\frac{1}{2}
 \end{array} \tag{5}$$

For larger n , $\mathcal{P}n$ will avoid such a trap in a match with TFT although the number of steps expected until absorption in the Cn state of mutual cooperation can be quite large. For example if TFT plays Pavlov starting with $\mathcal{P}(5, 10)$ it will require an average of 179 steps to train: see Figure 3.

Yet even for large n , $\mathcal{P}n$ can lock into a cycle with negative payoff against a vindictive version of TFT. The variant of TFT, which defects after a defection by the opponent and continues to do so until her opponent has cooperated twice in succession, will never train $\mathcal{P}3$ if he makes even one defection, or even if he was incorrectly perceived to have done so. Another variant, which requires at least $(n+1)/2$ sequential repentances (CD) by Pavlov after a single defection, will never get them from $\mathcal{P}n$. But against any reasonably forgiving strategy, $\mathcal{P}n$ is indeed trainable for $n > 2$.

A similar argument shows that even the more forgiving Tit-for-Two-Tats (TF2T) can get locked into such cycles with negative payoff to each in matches against $\mathcal{P}(0, 1)$ and $\mathcal{P}(0, 2)$. During such matches, $\mathcal{P}1$ would win an average of $1/4$ per round while TF2T would lose $3/4$. Similarly $\mathcal{P}2$ would win an average of $1/9$ while TF2T would lose an average of $7/9$. Thus these strategies exploit TF2T by almost 1 point per round. In Figure 3 we see that, for larger n , the training time for $\mathcal{P}n$ against TF2T will be only slightly shorter than against TFT, but the total payoff to TF2T is considerably less than that to the exploiting Pavlov.

4. PAVLOV AGAINST PAVLOV

Our greatest surprise in PPD is the result that $\mathcal{P}(k, 10)$ and his twin $\mathcal{P}(j, 10)$ will attain the stable state of mutual cooperation almost as rapidly as with $\mathcal{P}(k, 10)$ against the nearly ideal Normal Trainer strategy. We generalize here to other n . With strategy $\mathcal{P}(0, 1)$ against $\mathcal{P}(1, 1)$ the initial play will be DC which causes both players to change to the strategy $\mathcal{P}(0, 1)$ of certain defection. But after this mutual defection, each Pavlov will be punished and so each will cooperate the next round. So they arrive at the stable state $\mathcal{P}(1, 1)$ of mutual cooperation in 2 steps. During this training period the first player

would gain 1 unit while the second would lose 3. After that each player will continually win 1 unit each round. This is a small cost indeed.

In general in a match between Pavlovs starting with strategies $\mathcal{P}(k, n)$ and $\mathcal{P}(j, n)$, the lack of coordination of choices (CD or DC) will discourage cooperation by both parties and each will tend to degenerate quickly to the state of mutual defection with both playing $\mathcal{P}(0, n)$. From that stage on each will play with exactly the same probability. This emphatically does not mean that they will choose the same option but merely that the probabilities that they do so are the same. The remainder of the match is thus exactly the same as in a match between $\mathcal{P}n$ and NT.

For example if $n = 10$, among all initial strategies it takes longest to train the uncooperative $\mathcal{P}(0, 10)$ against the cooperative $\mathcal{P}(10, 10)$ (137 rounds). For comparison the training time for $\mathcal{P}(0, 10)$ against $\mathcal{P}(0, 10)$, or equivalently against NT, is 129 rounds, just a 6% difference. See Table 14 in PPD. The mutual training time for $\mathcal{P}n$ against $\mathcal{P}n$ with any initial probabilities is comparable to that of NT and thus is far faster than against TFT or TF2T (see Figure 10 in PPD and Figure 3).

As noted in Section 3, $\mathcal{P}n$ outscores the faster learning ($n > m$) $\mathcal{P}m$ in matches against All-C, All-D and Random from various initial starting probabilities. Indeed $\mathcal{P}n$ will outscore $\mathcal{P}m$ in head to head competition over many rounds. Figure 4 shows the expected payoffs in a match starting with strategies $\mathcal{P}(2, 3)$ against $\mathcal{P}(4, 6)$ for the first 20 rounds. Although $\mathcal{P}3$ will outscore $\mathcal{P}6$ by a very small amount over the first seven rounds, after that the slower reacting $\mathcal{P}6$ outperforms his faster learning cousin. The $\mathcal{P}3$ strategy overreacts to the various outcomes and tends to cooperate even though the $\mathcal{P}6$ strategy is still likely to defect. Figure 5 shows the training times for $\mathcal{P}(n - 2, n)$ against $\mathcal{P}(m - 2, m)$.

Among the Pavlovian strategies, only $\mathcal{P}1$ is deterministic. As noted before, Rapoport and Chammah (1965) called this the simpleton strategy. As we have seen above, the appellation is well deserved. Even $\mathcal{P}2$ does not fare well in many contests. It is not wise to be overly reactive. A bit more unpredictability is actually desirable. In other simulations, Bixenstine and Gaebelein (1971) considered still other stochastic variants of TFT. Their 'slow-slow' variant has a similar

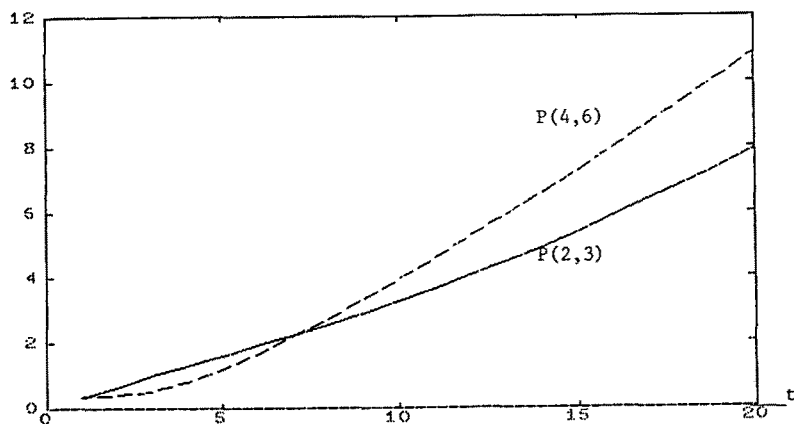


Fig. 4. $P(2,3)$ versus $P(4,6)$. Payoffs as a function of the number of rounds, t .

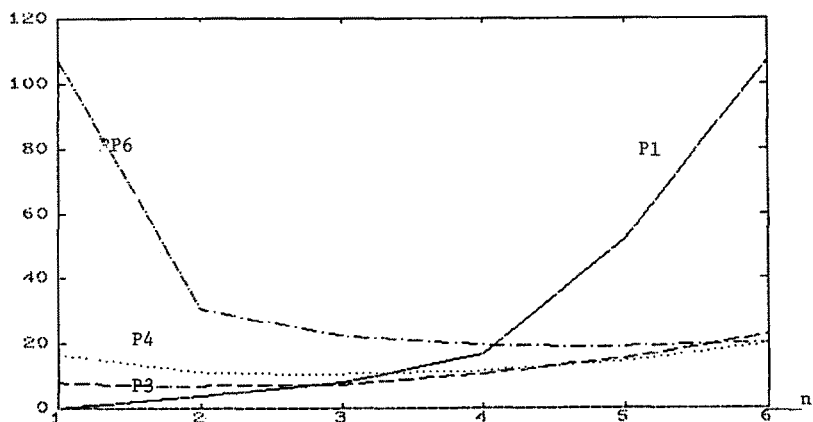


Fig. 5. $P(k-2, k)$ versus $P(n-2, n)$ for $k = 1, 3, 4$, and 6 . Training times as a function of n .

relation to TFT as our $\mathcal{P}5$ does to $\mathcal{P}1$. They found that this slower learning variant evoked higher cooperation than the original version.

For most opponents of Pavlov that we have considered, the transition matrix has a single absorbing state corresponding to cooperation with certainty. In other words there is a chain of events leading from any state to the state of mutual cooperation and that thereafter both players remain in that state. The following proposition follows immedi-

ately from the definitions and the above arguments. In particular it implies that the strategies $\mathcal{P}n$, for $n = 1, 2, \dots$, form a clan.

PROPOSITION 6. *If the state of mutual cooperation is the only absorbing state in a Markov matrix $\mathbb{A}(\mathcal{S}, \mathcal{T})$ between strategies $\mathcal{S}$ and $\mathcal{T}$, then these two strategies are fraternal.*

If n is large and encounters are frequent with payoff matrix

$$\begin{array}{|cc|} \hline 1, 1 & -2, 2 \\ 2, -2 & -1, -1 \\ \hline \end{array},$$

then the training time for two Pavlovs each using initial strategies $\mathcal{P}(n/2, n)$ increases exponentially in n , as Figure 10 of PPD illustrates. The Pavlovs will tend to defect with high frequency for extended numbers of rounds since the downward movement after a CD or DC outcome is twice as great as the upward movement after a CC or DD outcome (see Tables 16 and 17 in PPD). Equivalently, if n is moderate but the payoffs are very small, then the players will tend to defect until a rare sequence of CC outcomes boosts their probability of cooperation to the absorbing state. If real life opponents are petty or jealous in small things, so that the effective n is large, then they will tend to degenerate toward this somewhat uncooperative state and remain there for many steps. There may be some mathematical validity to Henry Kissinger's quip: The battles in academia are so fierce because the stakes are so low.

We now address the question: For which k and n is $\mathcal{P}(k, n)$ the best initial Pavlovian strategy? Our first criterion in Definition 1 for an ideal strategy is that it should collect a large sum over many rounds against a range of opponents. No strategy can average more than $+1$ per round over long runs unless the opponent is incredibly self-sacrificing such as All-C. Results from computer tournaments and other simulations suggest that a good strategy should start out being nice. This suggests that for each n , the best strategy is the cooperative $\mathcal{P}(n, n)$. Against All-D, this initially nice strategy will rapidly approach the equilibrium probability vector for moderate n . In other words Pavlov does not fare much worse by starting out cooperative.

Our third criterion is that it should be asymptotically cooperative. It should not be overly disturbed by a rare defection, real or perceived. The ideal strategy recovers the state of mutual cooperation quickly, at least with its clone and other members of the clan. If in one round between cooperating Pavlovian strategies an outcome was miscommunicated or wrongly given, then the resulting CD or DC outcome will drop the cooperating $\mathcal{P}(n, n)$ strategy to the state $\mathcal{P}(n-2, n)$ no matter which player was perceived to have defected. If the opponent is $\mathcal{P}(m, m)$, he would simultaneously have fallen to $\mathcal{P}(m-2, m)$.

Random noise could also spontaneously affect one player's initial cooperation probability, marginally. Factors, such as the weather, may make one happy or grouchy. Even a trained Pavlov might have a bad day and shift to $\mathcal{P}(n-1, n)$. For these reasons, we should not restrict our attention only to nice Pavlovian strategies, even in matches against nice opponents. On the other hand, we lose little in concentrating on initial strategies which tend to cooperate, and in particular, on $\mathcal{P}(n-2, n)$.

As previously noted, the larger the value of n , the more likely that $\mathcal{P}n$ will outscore All-C, Random and even $\mathcal{P}m$. On the other hand, the training times in matches between $\mathcal{P}(n-2, n)$ and TFT, PT or $\mathcal{P}(m-2, m)$ grow rapidly with n (see Figures 3a and b). The expected average payoff during this training period will usually be quite low to

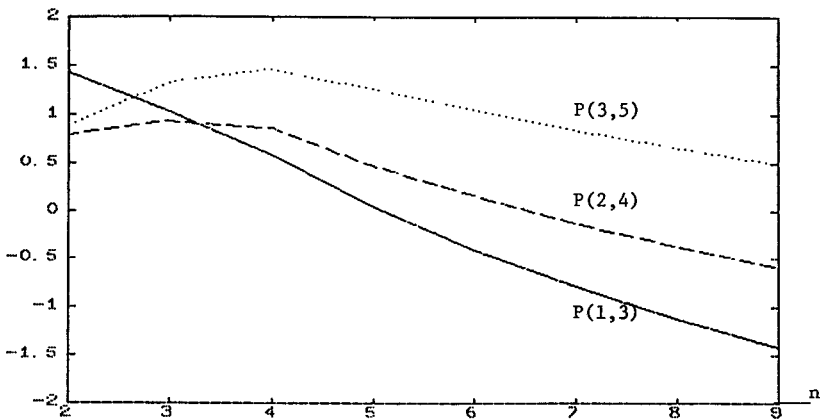
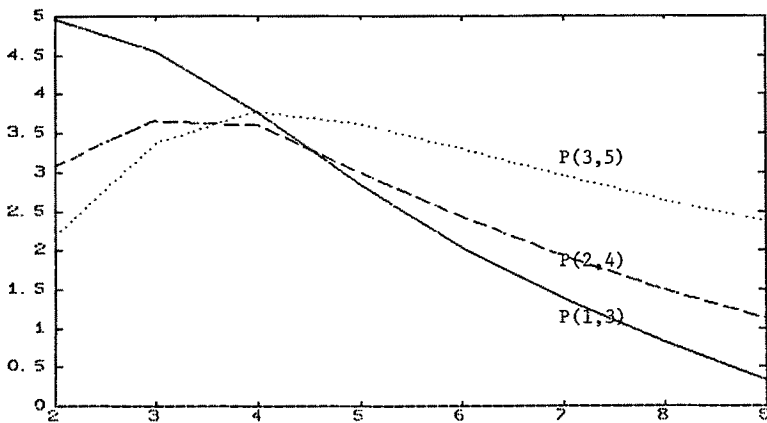
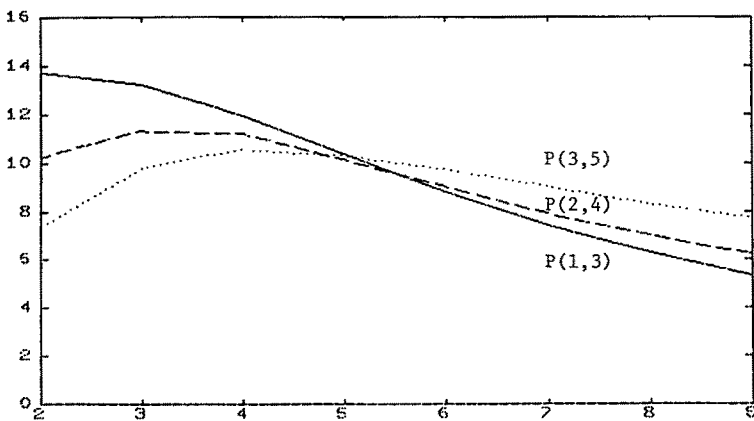


Fig. 6a. Average payoff to $P(1, 3)$, $P(2, 4)$ and $P(3, 5)$ versus $P(n-2, n)$ as a function of n for $w = 5/6$.

both players. Thus if n is large the first criterion is violated. In a noisy environment, the initial strategy $\mathcal{P}(n-2, n)$ should be studied as well as the nice version $\mathcal{P}(n, n)$ for reasons discussed above. This strategy collects a higher total score against its clone and other Pavlovian opponents when n is about 3 or 4 (see Figure 6). Note that $\mathcal{P}(0, 1)$ and $\mathcal{P}(0, 2)$ do not perform well against other Pavlovs. These strategies could also end up in endless negative payoff cycles against TFT and TF2T, so we rule out $\mathcal{P}1$ and $\mathcal{P}2$ as candidates for optimal strategies.

Fig. 6b. $w = 10/11$.Fig. 6c. $w = 20/21$.

The optimal value of n will depend not only on the opponent but also on the expected number of encounters. No iterated contest can go on forever and slow learning strategies may not have a chance to build up their lead in a few matches. If we assume in advance a fixed number of rounds of a match, then one can easily prove that the All-D strategy is the only 'rational' strategy and so both players will lose. Alternatively and more realistically, following Shubik (1970), one can model a match that lasts an average of t plays by postulating a certain probability w that the two players will meet, or having met will meet again. For example if $w = 0.5$, then with 50% probability they will meet once, with 25% probability they will meet twice, etc. The expected number of encounters is $w + w^2 + w^3 + \dots = w/(1 - w)$ or just once with $w = \frac{1}{2}$. Note that if the expected number of meetings is $t = w/(1 - w)$ then $w = t/(1 + t)$.

TABLE III

All-D versus $\mathcal{P}(n, n)$. The fixed and expected number of rounds that All-D outscores a nice strategy for $n = 1$ to 20.

n	t	w	$\frac{w}{1 - w}$
1.0000	1.0000	0.5000	1.0000
2.0000	1.0000	0.4140	0.7065
3.0000	1.0000	0.5270	1.1142
4.0000	2.0000	0.5770	1.3641
5.0000	3.0000	0.6310	1.7100
6.0000	3.0000	0.6680	2.0120
7.0000	4.0000	0.7010	2.3445
8.0000	4.0000	0.7280	2.6765
9.0000	5.0000	0.7500	3.0000
10.0000	5.0000	0.7690	3.3290
11.0000	6.0000	0.7850	3.6512
12.0000	6.0000	0.8000	4.0000
13.0000	7.0000	0.8120	4.3191
14.0000	7.0000	0.8230	4.6497
15.0000	8.0000	0.8330	4.9880
16.0000	9.0000	0.8420	5.3291
17.0000	9.0000	0.8500	5.6667
18.0000	10.0000	0.8570	5.9930
19.0000	10.0000	0.8630	6.2993
20.0000	11.0000	0.8690	6.6336

As we have seen in Section 3, the Perfect Trainer can get away with a defection against a trained Pavlov only if n is very large or if they will likely not meet often. Thus the All-D strategy can outscore PT against Pavlov in the short run. We can estimate the break-even point in a match between $\mathcal{P}n$ and All-D in two ways. We might consider a match with exactly t rounds or we might postulate a discount parameter w which will in turn correspond to an expected number of rounds, $w/(1-w)$, in the match. In each case the break-even point depends on the value of n . Table III and Figure 7 shows that All-D would score higher than a nice strategy against $\mathcal{P}(n, n)$ for roughly the first $n/2$ rounds while, using the discount parameter method, All-D would be the better strategy only in matches with expected lengths of $n/3$ rounds.

The results above again indicate that n should not be large. In Figure 8 we note that the deficit to All-D peaks when n is close to 3 or 4 for various values of w . This means that All-D would not long thrive against $\mathcal{P}(3, 3)$ or $\mathcal{P}(4, 4)$. In Figures 6a, b, and c, we plot the average payoff to $\mathcal{P}(1, 3)$ and $\mathcal{P}(2, 4)$ and $\mathcal{P}(3, 5)$ in matches against $\mathcal{P}(n-2, n)$ for $n = 2, \dots, 9$ for matches with discount parameters corre-

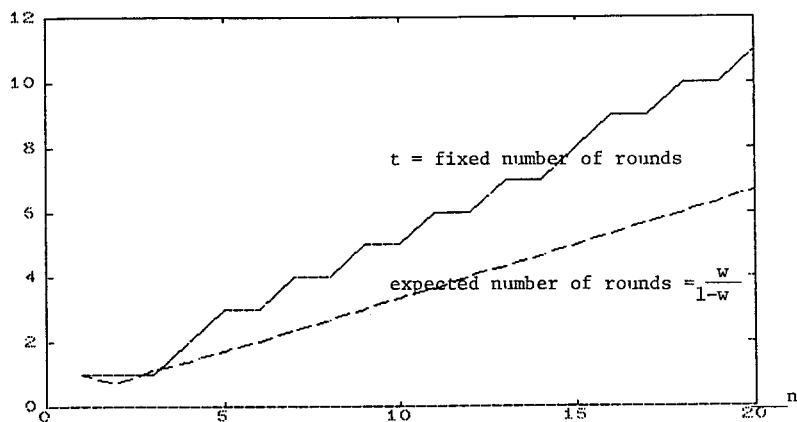


Fig. 7. All-D versus $\mathcal{P}(n, n)$. Fixed number of rounds (t) and expected number of rounds $w/(1-w)$ for discount parameter w that All-D can outscore a nice opponent, as a function of n .

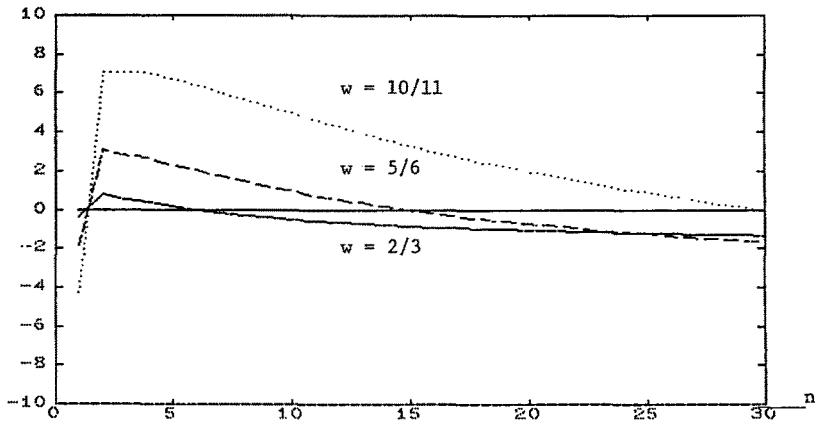


Fig. 8. All-D versus $P(n, n)$. Deficit to All-D as a function of n for various values of the discount parameter w .

sponding to 5, 10 and 20 expected encounters. Again the optimal performance occurs when $n = 3$ or 4 for matches of moderate length.

5. APPLICATIONS AND SIMULATIONS

Many real life encounters have been heuristically, albeit simplistically, modeled by the iterated Prisoner's Dilemma game. In this section we briefly sketch some 'real world' examples and a few human simulations. Several authors have proposed mechanisms by which players of many species in a variety of games such as those below have adopted the TFT strategy to encourage mutual cooperation. We claim that the Pavlovian strategies account for the actual responses as well or better than TFT.

Axelrod (1984) discusses the strange truces that often took place across some trench lines in World War I. French and German soldiers would refrain from shelling each other or, if they shelled, they took care to do no injury to the others (CC) much to the chagrin of their senior officers. Their reasoning was that if they mauled the enemy (DC), then the other side would retaliate, perhaps with more intensity (CD). The officers occasionally managed to restore 'normal warfare' by rotating their troops (decreasing the discount parameter) and

sending them out on raids and forays (DD with smaller penalty payoff P).

Lombardo (1985) has modeled behavior by tree swallows using the Prisoner's Dilemma. Cooperation means peaceful coexistence between parents and non-breeding swallows while defection means attacking the other birds or the fledglings. Each type of bird will normally cooperate. When Lombardo simulated defection by the non-breeders (he put dead fledglings in the parents' nest) the parents retaliated, often with considerable violence. After a few encounters peaceful coexistence was again the norm.

International trade has often been modeled by the Prisoner's Dilemma. The cooperative strategy would be to lower tariffs, thereby increasing wealth and commerce, while the defection option would be to raise tariffs to promote domestic industries. If both countries cooperate then each side profits. If one side raises tariffs, then its industries benefit to the great expense of the other side, provided that the other country does not also raise its tariffs. If both sides raise tariffs then the resulting trade war stifles both economies and both suffer. Historical examples confirm many instances of retaliation for defection and also suggest the eventual restoration of freer trade (Conybeare, 1987).

In each of these three cases, and others in the literature (see especially the annotated bibliography of Axelrod and Dion, 1987), the Prisoner's Dilemma seems to be an appropriate, if somewhat superficial, model. Each side benefits a bit by mutual cooperation but one side could benefit much more if it defects and the other cooperates. The 'rational' solution of mutual defection is not good to either. Yet somehow mutual cooperation usually tends to evolve. And if a defection does occur in any of the examples above, it is typically met with swift retaliation. So we see that CC is usually followed by more cooperation and a defection (CD) by retaliation.

Both the TFT and the Pavlov strategies account for these responses so far. But how do these strategies react if they themselves were first to defect, or were even perceived to defect? TFT would see the cooperation of her opponent and cooperate herself on the next round. Pavlov would reap the temptation reward and so tend to defect again, exploiting his opponent. How would the German troops have acted if

their initial shelling brought a feeble response from the French? Various studies (see Patchen (1986) and references therein) have indicated that exploitation generally continues in the absence of retaliation.

Assume again that one side defects or was perceived to have defected. If retaliation is swift then the original defector has no time to savor the fruits of his deception in the cases above. For example, the advantages that domestic industries would gain by erecting stiff trade barriers unilaterally may take months or even years to become significant. Thus rapid retaliation in response to a defection may more closely resemble a DD outcome than a DC followed by CD. If either or both players used the TFT strategy, one would expect the struggle to escalate (further DD outcomes). On the other hand if the players were using the Pavlov strategy, then gradually, but not monotonically, the outcomes would evolve back towards mutual cooperation, as apparently happens in the examples.

The expected time for retraining depends exponentially on the magnitude of the payoffs and thus on the learning rate of the participants. The French and German officers may have effectively lowered the learning rate of their troops by ordering raids which did relatively minor damage. That combined with frequent rotation of troops made it less likely that mutual cooperation would ever resume.

In PPD we showed that if two individuals initially used strategy $\mathcal{P}(5, 10)$, then they would learn to cooperate (reach the absorbing state) after an average of 110 plays. The training time in the Pavlov versus Pavlov model is roughly consistent with some student simulations of the Prisoner's Dilemma run by Rapoport and Chammah (1965). In those simulations, students would cooperate or defect more or less at random for the first few plays while learning the game and then they would each defect more and more often. Occasionally one would try to break out with a C but this would set them back unless the opponent simultaneously cooperated. But eventually such CC outcomes did occur. From about round 50 to round 150 the frequency of CC outcomes increased from about 25% to 65%. This might signify that more and more of the pairs have become trained, i.e., locked into the absorbing CC state.

The predicted frequency of CC outcomes in a match between

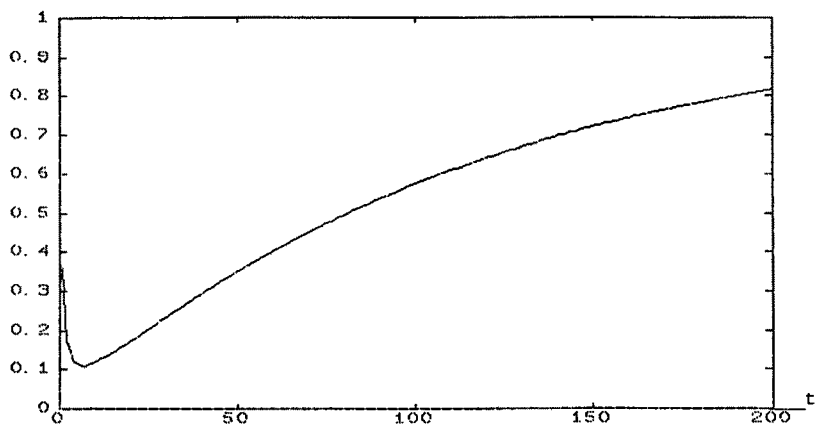


Fig. 9a. $P(4, 10)$ versus $P(9, 10)$. Predicted frequency of CC outcomes as a function of the number of rounds played.

$\mathcal{P}(4, 10)$ against $\mathcal{P}(9, 10)$ is given in Figure 9a. It compares favorably with the chart on p. 180 of Rapoport and Chammah, which is reproduced in Figure 9b.

We now feel that learning must take place much more rapidly if it is to be an effective strategy in a real life interaction. In other words we feel that n is closer to 3 or 4 than to 10. Perhaps the payoffs of 1/10 cent per point in the Rapoport–Chammah simulation discussed above were not sufficiently valued by the student players. Perhaps the participants felt that the experiment was ‘only’ a game. Students who were used to zero-sum games such as poker, chess, and sports, may have tailored their strategy in the Prisoner’s Dilemma simulation toward beating the opponent rather than doing well for themselves, regardless of the instructions.

Opp (1988) reported on an interesting simulation with a payoff of far greater utility to the subjects than the 1/10 cent per unit payoff in the experiment above. Students were presented with the following scenario: They and their neighbor each played their stereo at the same time each day. The cooperation strategy would be for both to play their stereos softly. If one turned down his volume but his neighbor played hers as loudly as she wished to, then the cooperator would be disturbed. If both played the stereo loudly then they could listen but

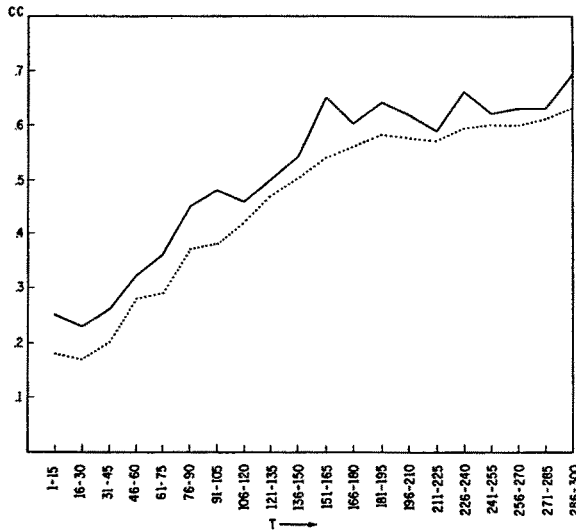


Figure 34. Comparison of observed time course of CC in the Pure Matrix Condition (solid line) with that obtained from a simulated Markov model (dotted line) with absorbing states. Parameter values: $\alpha = .6$; $\beta = .43$; $\gamma = .35$; $\omega = .30$; $\gamma = .018$; $\delta = .0024$.

Fig. 9b. From *Prisoner's Dilemma*, Rapoport and Chammah, p. 177, University of Michigan Press, Ann Arbor, 1965.

with mild discomfort. Over a simulation of 10 encounters with each of the 26 pairs of students, Opp observed that the students' behavior was not in accord with that predicted by the TFT strategy. In particular students exploited their cooperative partners (a DC outcome was followed by another exploiting D) 99 times out of 158. Also retaliation did not seem to increase the likelihood of cooperation in the near term. It is not clear from Opp's article how students reacted to the DD outcome but we feel that the evidence is more supportive of the Pavlov than the TFT hypothesis.

In another interesting simulation Komorita, Hilty and Parks (1991) had 96 students play 36 rounds of a standard iterated Prisoner's Dilemma situation against TFT and variations which are slower to reciprocate cooperation and/or defection. The students were told they were playing other students. Apparently students rarely became 'trained' to cooperate in the matches against any of the variants (their

average play of C was only about 50%). Our Markov chain results suggest that the TFT strategy usually requires many more than 36 rounds to complete training. We suggest that these students would have achieved a higher level of cooperation in those 36 rounds if they had played against a programmed $\mathcal{P}(3, 3)$ or $\mathcal{P}(4, 4)$ strategy.

A computer simulation of the iterated Prisoner's Dilemma with noise by Bendor *et al.* (1991) confirmed that the level of cooperation, or generosity, of their more successful strategies must adapt to the conditions at hand. The rigid TFT strategy performed poorly in their tournament as she did in the one of Donniger (1986). The best strategies reacted, as does our Normal Trainer, to the long term perceived level of cooperation of the opponent in the simulated noisy environment. But how best to determine the correct level of cooperation? In the presence of noise, as Axelrod and Dion (1988) have said "there is a trade-off: unnecessary conflict can be avoided by generosity, but generosity invites exploitation." The Pavlov adapts his level of cooperation by reacting not to the play of the opponent but to the rewards and punishments that he receives. He will retaliate against mean opponents and exploit overly generous opponents as well as (eventually) cooperate with those with more balanced strategies.

6. CONCLUDING REMARKS

How close can one come to an ideal strategy for the iterated Prisoner's Dilemma? To score a high average, computer simulations suggest that the ideal strategy should not be the first to defect. Nice strategies frequently outperform all others, at least if the discount parameter is not very low. But strategies that are too tolerant, such as TF2T, can be exploited. On the other hand while All-D and the Suspicious TFT strategies can never be exploited, they fare poorly in overall winnings even against their own clones, so we reject them as serious contenders for ideal strategies. TFT is a nice strategy which can never be seriously exploited and many researchers believe it to be the ideal strategy. Yet TFT in turn performs miserably in a noisy environment against its clone.

The Pavlovian strategies will average a high score against all nice opponents and will even exploit overly nice ones. They in turn can be

exploited but only by very nasty strategies such as All-D and then, if the discount parameter is even moderately large, the nasty strategy will not do as well as a nice strategy against Pavlov. In addition, $\mathcal{P}_n$ is stable against noise in a match with most strategies, at least for $n > 2$. For these and other reasons, we propose that the strategy $\mathcal{P}(n, n)$ is close to ideal. Since the training time grows exponentially in n , our candidate for nearly ideal strategy is $\mathcal{P}(3, 3)$ or $\mathcal{P}(4, 4)$ depending on the type of opponents and the expected length of the match.

Simon (1991) argues that bounded rationality as well as social learning are keys to the development of altruism. Indeed our Pavlov is a perfect example of a player who does not reason but only reacts reflexively. A smarter player might understand the nature of the payoff matrix and choose the 'rational' All-D strategy or at least try some probing defections. A player with still more intelligence would recognize the desirability of the iterated CC outcome and adopt a TFT or similar strategy, not quite aware of the danger of being trapped in retaliatory cycles. A truly proficient reasoner, such as NT, would realize not only that there is no long term gain in trying to cheat but also that mechanical retaliation can be self destructive against some strategies.

We would call the Normal Trainer the far-sighted strategy. In a game with many rounds, this strategy would collect high total points against a wide range of opponents. The All-D strategy is the short-sighted strategy. It will perform well for a few rounds at most. Following this pattern, we might call Pavlov the hind-sighted strategy. It learns how to play its opponents not by analysis but by reaction. Yet not only do we feel that this strategy is better than TFT in most situations, it can even be better than NT. If the opponent were particularly unreasoning (for example All-C or Random) then Pavlov will exploit this weakling and do better than NT would.

A strong case can be made that the Pavlovian strategy is the natural strategy in the animal world. Many animals, certainly higher mammals, do indeed learn by Pavlovian conditioning. The advantages of such conditioning go far beyond mere two person Prisoner's Dilemma type games to the very survival of the species in changing environments, i.e., to 'games against nature'. The animals would transfer this innate adaptiveness to inter- and intraspecies interactions of the Prisoner's Dilemma type as well as other two-person games such as Chicken.

Finally we argue that the Pavlovian strategy is only common sense, i.e., we have come to learn of its effectiveness in early life. People do tend to cooperate with friends and to retaliate if they feel they have been wronged. Many of us are greedy and will exploit others if we feel we can get away with it. And if we are trapped in a vicious cycle of mutual defections, then we will try to break out of it rather than continually absorb the negative payoffs. Our reactions are not completely predictable and it may take several positive reinforcements for us to be persuaded to cooperate with more certainty as well as several negative conditionings to sour us on the world. These reactions are all consistent with the $\mathcal{P}n$ strategy.

The Pavlovian strategy is not quite ideal. It can be exploited and it will exploit. It is neither envious nor altruistic. The strategy will not analyze its opponent or even itself. It reacts rather than reasons. Yet, remarkably, it works.

APPENDICES

A. Catalog of Strategies

$\mathcal{P}(k, n)$	Pavlov strategy with learning step $1/n$ and initial probability k/n of cooperation.	
All-D	Always defect	$p = 0$.
All-C	Always cooperate	$p = 1$.
$\mathcal{C}p$	Random strategy: cooperate with probability p .	
TFT	Tit-for-Tat.	
TF2T	Tit-for-2-Tats.	
PT	Perfect trainer: If opponent is more likely to Cooperate, then play C. Otherwise play D.	
NT	Normal trainer: Approximate but more realistic version of PT.	
$\mathcal{E}(k, n)$	Evolutionary strategy of n individuals k of whom cooperate.	
$\mathcal{EF}$	Error Forgiving	see Molander (1986).
$\mathcal{CC}$	Conditional Cooperating	see Mueller (1987).
$\mathcal{NS}(y, p, q)$	Reactive Strategy	see Nowak-Sigmund (1989).
Downing and C-Downing	See Axelrod (1984) and Donninger (1986).	

B. Markov Methods

In this appendix we present technical details about the Markov chain $\mathbb{A}(\mathcal{P}n, \mathcal{S})$ for various strategies $\mathcal{S}$. Our basic reference for terminology, theorems, and proofs is Kemeny–Snell (1976). We first describe how to determine the number of states (the order of the Markov matrix) and the transition probabilities (the matrix entries). Each state corresponds to combinations of the possible mixed strategies for each player. Pavlov has exactly $n + 1$ mixed strategies or states, $\mathcal{P}(k, n)$ for $k = 0, \dots, n$. The probability, p , that strategy $\mathcal{S}$ will cooperate on the next round may depend on many factors.

For some strategies $\mathcal{S}$, p is fixed once and for all. If $\mathcal{S}$ is the All-D, Random, or All-C strategy, then p is 0, $\frac{1}{2}$, or 1 respectively. For other strategies the probability may depend only on the current state of Pavlov. If $\mathcal{S}$ is the Perfect Trainer, then $p = 0$ if $k/n \leq \frac{1}{2}$ and $p = 1$ otherwise. If $\mathcal{S}$ is the Normal Trainer then $p = k/n$; NT will cooperate with precisely the same probability as her opponent. In each of the cases above, the Markov chain has exactly $n + 1$ states.

If $\mathcal{S}$ is TFT then p depends not only on the current state but also on the previous play; it will be 0 or 1 depending on whether $\mathcal{P}n$ defected or cooperated on the previous round. This Markov chain has $2(n + 1)$ state which we list as $D(0/n), C(0/n), D(1/n), \dots, C(n/n)$. For example $D(3/4)$ means that Pavlov defected on the previous round and has a 75% chance of cooperating on the next round. Since TFT will defect ($p = 0$), the only possible outcomes are CD with 25% probability, transforming down two increments to state $C(1/4)$, and DD with 75% probability, transforming up one increment to state $D1$.

If $\mathcal{S}$ is $\mathcal{P}m$, then the strategies of each player are, to some extent, independent. This Markov chain has $(n + 1)(m + 1)$ states which we index by $(k/n, j/m)$. After a CC or DD outcome, each numerator is increased by 1 and after a CD or DC outcome each is decreased by 2, subject to the usual boundary constraints. Note that the difference in numerators, $k - j$, is usually preserved. It can change only along the edges since the probabilities for cooperation must be between 0 and 1. For example, state $(3/n, 1/m)$ will transform either to state $(4/n, 2/m)$, preserving the numerator difference, or to the edge state $(1/n, 0/m)$. Similarly a CC outcome from state $(k/n, m/m)$ will cause a transition to state $[(k + 1)/n, m/m]$. The numerator difference will decrease if $k < m$ and increase if $n > k > m$.

transition to state $[(k+1)/n, m/m]$. The numerator difference will decrease if $k < m$ and increase if $n > k > m$.

If $m = n$ then the subset of ‘diagonal’ states of the form $(k/n, k/n)$ is an absorbing set in the sense that once in one of these states, the strategies of the two Pavlovs will ever after be identical. They move in tandem, erratically, toward mutual cooperation. The Markov chain corresponding to this subset of diagonal states is precisely the same as in the Markov chain for Pavlov against the Normal Trainer.

The Markov matrix yields a great deal of information about the evolution of cooperation. The matrices for Pavlov against All-D and Random are *regular*; any state can be reached from any other state after sufficiently many transitions. For such matrices, the rows of the matrix $\mathbb{A}^t$ each tend to a unique probability row vector U as t increases. This vector satisfies $U\mathbb{A} = U$; U is the left eigenvector with eigenvalue 1 such that the entries are all positive and add up to 1. See Section 4.1 of Kemeny–Snell (1976). Cooperation will never become the rule with these strategies.

Assume that there is a positive probability for transforming from any state to the mutual cooperation state after a sufficiently large number of steps and that once in that state one can never leave. In this case the Markov chain is called *absorbing* and the mutual cooperation strategy is the unique absorbing state. The Markov matrix $\mathbb{A}$ for such chains can be written in block form as

$$\begin{bmatrix} Q & R \\ 0 & 1 \end{bmatrix}$$

where Q is the non-singular submatrix of states other than mutual cooperation. The matrix

$$N = (I - Q)^{-1} = I + Q + Q^2 + \cdots \quad (\text{A1})$$

is called the *fundamental* matrix of $\mathbb{A}$. Its entries ν_{ij} measure the frequency of visits to state j starting from state i . Thus the *training time*, or time to absorption, starting from state $\mathcal{P}(i, n)$ is the sum

$$q = \sum_{j=0}^{n-1} \nu_{i,j} = N_i \mathbf{1} \quad (\text{A2})$$

where N_i is the i th row vector of N and $\mathbf{1}$ is the column vector of 1's. See Section 3.5 of Kemeny–Snell (1976) for more details. The MATLAB programs to find the expected training time for various strategies are available on request.

If Pavlov plays All-C then the Markov chain has two absorbing states, mutual cooperation, $\mathcal{P}(n, n)$, and continual exploitation, $\mathcal{P}(0, n)$. The theory above extends to giving the probability of absorption in one or the other state. This probability can be read off of the $n \times 2$ matrix NR (see Theorem 3.3.7 of Kemeny–Snell (1976) and Figure 5 of PPD). In particular if Pavlov starts in state $\mathcal{P}(2, 4)$ then a MATLAB computation establishes that there is a 61% chance that Pavlov will continue to exploit All-C.

C. Long Term Behavior

We can estimate the expected payoff to Pavlov and an arbitrary opponent in exactly t rounds as follows. The (i, j) th entry of $\mathbb{A}$ represents the probability, or the expected frequency, that $\mathcal{P}(j, n)$ will be the second strategy given that $\mathcal{P}(i, n)$ was the first strategy. Similarly the (i, j) th entry of $\mathbb{A}^t$ represents the expected frequency that $\mathcal{P}(j, n)$ will be the strategy used on round $t + 1$ given that $\mathcal{P}(i, n)$ was the first strategy. Thus the expected frequency in any state in the first t rounds is determined by the entries in the matrix

$$\mathbb{F}_t = I + \mathbb{A} + \mathbb{A}^2 + \cdots + \mathbb{A}^{t-1}. \quad (\text{A3})$$

Since each of the above summands are Markov transition matrices with row sum equal to 1, each row sum of $\mathbb{F}_t$ must be t . The expected payoff to $\mathcal{S}$ in those t rounds is given by the $(n + 1)$ column vector $\mathbb{F}_t \mathbb{S}$. For example if Pavlov uses strategy $\mathcal{P}(0, n)$ initially, then the expected payoff to $\mathcal{S}$ is the first component of that vector.

Consider now a match between $\mathcal{P}(n, n)$ and the cheating Perfect Trainer as in Section 3. If PT had continued to play her strategy of cooperating with cooperative opponents, then after t rounds each would have collected t units. Assume however that PT defects on the first round, collects 2 rather than 1, and thereby drops Pavlov to strategy $\mathcal{P}(n - 2, n)$. Let V be the $(n - 1)$ row vector of $\mathbb{F}_t$, corre-

sponding to the new initial Pavlov state $\mathcal{P}(n-2, n)$. After t more rounds, PT would have collected a total of $2 + V \cdot \mathbb{S}$.

If this number is less than $t + 1$, then PT would have been better off not cheating. By Definition 4, the surplus (deficit) is the amount that the cheater will win (lose) relative to the pure Perfect Trainer strategy. For $t = 1$ this surplus is 1, i.e., the cheating PT gains initially. If $n = 10$, then on the second round, formula (2) gives the expected payoff from state $\mathcal{P}(n-2, n)$ to PT is $1 - 3(2/10) = 0.4$ so the surplus is $2.4 - 2 = 0.4$. The 9th row vector of $\mathbb{F}_3 = I + \mathbb{A} + \mathbb{A}^2$ is

$$V = [0, 0, 0, 0, 0.08, 0, 0.2, 0.2, 1.0, 0.8, 0.72] \quad (\text{A4})$$

so the expected payoff in the three rounds after the cheat is $V \cdot \mathbb{S} = 1.676$ where

$$\mathbb{S} = [-1, -0.7, -0.4, -0.1, 0.2, 0.5, \\ -0.2, 0.1, 0.4, 0.7, 1.0]^T$$

is the payoff column vector for PT (see (2) and PPD). In the four rounds, the cheating PT has collected $2 + 1.676$ units, whereas she would have had 4 units had she not defected, for a deficit of approximately 0.334.

This argument extends to find the number of rounds for which All-D can expect to stay ahead of a nice opponent when playing Pavlov who starts in the cooperative state $\mathcal{P}(n, n)$. Let W be the bottom, or $n + 1$, row vector of $\mathbb{F}_t$. Then All-D can expect to collect $W \cdot \mathbb{S}$ units in t rounds, where $\mathbb{S} = [-1, -1 + 3/n, \dots, 2]$ is the payoff vector to All-D. As long as this entry is larger than t , All-D will collect more than the nice PT strategy would have; she will have a surplus. In Figure 7, the greatest t for which the surplus is positive for various values of n is found and plotted by a MATLAB program.

Unlike All-D, PT can get away with a cheat forever, provided n is very large. Write the Markov matrix

$$\mathbb{A} = \begin{bmatrix} Q & R \\ 0 & 1 \end{bmatrix}$$

where Q is a nonsingular $n \times n$ matrix and R is an $n \times 1$ matrix. Then

$$\mathbb{F}_t = \begin{bmatrix} Q(t) & M(t) \\ 0 & t \end{bmatrix}$$

$$\text{where } Q(t) = I + Q + \cdots + Q^{t-1} = N - Q^t N \quad (\text{A5})$$

and $M(t)$ is the column vector with n components determined by the condition that each row of $\mathbb{F}_t$ has sum equal to t . For t very large, we can approximate $Q(t)$ by N since Q^t approaches 0 (Theorem 3.1.1, Kemeny–Snell). Thus the $n - 1$ st component of $M(t)$ is approximately $t - N_{n-1}\mathbf{1} = t - q$ where q is the training time (see [A2]) for Pavlov starting in state $\mathcal{P}(n-2, n)$.

Note that the $n + 1$ (last) entry in $\mathbb{S}$ is 1, i.e., the payoff to PT is 1 if Pavlov, and so also PT, cooperates with certainty. Let $\mathbb{S}'$ be the column vector consisting of the first n components of the payoff vector $\mathbb{S}$ to PT. The expected payoff to PT after a large number, t , of rounds is approximately $2 + N_{n-1}\mathbb{S}' + t - q$ units and the surplus after the total of $t + 1$ rounds is thus

$$(2 + N_{n-1}\mathbb{S}' + t - q) - (t + 1) = N_{n-1}(\mathbb{S}' - \mathbf{1}) - 1. \quad (\text{A6})$$

In other words for the surplus to always be positive, we must have $N_{n-1}(\mathbb{S}' - \mathbf{1}) \geq 1$.

Using MATLAB we have computed these values for various values of t when $\mathcal{S}$ is the Perfect Trainer. In Figure 1 we see that n must be at least 31 for PT to keep winning. But even if n is so big, the long-term prospects for the cheating PT are poor. PT would have to wait about 500 steps, the training time for PT against $\mathcal{P}(29, 31)$, before having a reasonable expectation for getting away with another defection.

To model encounters with a discounted future payment, assume that the players will meet or, having met will meet again, with probability $w < 1$. Then the expected frequency in state $\mathcal{P}(j, n)$ given that the initial state was $\mathcal{P}(i, n)$, is the (i, j) entry in the matrix defined by the convergent infinite series of matrices:

$$\mathbb{B}_w = wI + w^2\mathbb{A} + w^3\mathbb{A}^2 + \cdots = w(I - w\mathbb{A})^{-1} \quad (\text{A7})$$

For $\mathcal{S} = \text{All-D}$, it is possible to compute $\mathbb{B}_w$ explicitly in terms of w and to show that All-D will do worse against $\mathcal{P}(k, 2)$ than would a nice

strategy even if they are unlikely to meet even once more. For large n it is not feasible to compute $(I - wA)^{-1}$ symbolically. Instead we use MATLAB to home in on the break-even point of $\mathcal{P}(n, n)$ against All-D. See Table III.

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Department of Mathematics,
Duke University, Durham NC 27708, U.S.A.

Department of Mathematics,
Meredith College, Raleigh NC 27606, U.S.A.